

EUGENE L. BUCHANAN

Quality, Test & Controls Engineer — Solar Thermal, Waste-to-Energy & Embedded Systems

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San Quintín, BC, MX | +52 (616) 126 5089 | Relocating to Austin, TX or Miami, FL · Available immediately

Target roles: Quality Engineer · Reliability Engineer · Lead Controls & Firmware Engineer · Technical Program Manager, Supply Chain & Manufacturing

SUMMARY

Quality and controls engineer with 30 years across test engineering, product quality ownership, and thermal-energy systems. Twenty-two years in solar and renewable energy, where I produce permit-ready CAD packages — plot plans, single-line electrical diagrams, string impedance calculations, and NEC 690 compliance and safety calculations — coordinate engineer stamping, and run them through city plan review to permit to operate, and design custom mechanical parts where a catalog item does not fit. Sold and integrated more than \$32M in solar, wind, and geothermal air-conditioning installations. Process research on Fischer-Tropsch syngas stabilization, hot gas flow control, and anti-fouling. Ten years of test ownership spanning embedded device hardware, high-availability server hardware, and production software — from device-level debugging on ARM/EPOC through WHQL certification testing of RAID clusters under sustained high-load traffic. Built a test automation function from nothing and drove a 53% reduction in operating expenditures on a \$16,027,777 annual public-agency budget. A.S. Mechanical Engineering, 1996.

CORE COMPETENCIES

Quality & Test Engineering — quality management · quality standards · quality assurance · test planning · test cases · inspection · calibration · root cause analysis · 5-Why · defect triage · white-box testing · end-to-end regression · load and performance testing · security and usability testing · quality metrics · technical documentation · blue-team incident forensics

Embedded & Controls — C · C++ · ARM · embedded Linux · device flashing · off-target crash and memory-log capture · on-device debugging · SCADA · instrumentation · sensors and relays · VFD and motor control

Solar Design & Permitting — photovoltaic and solar thermal system design · off-grid and on-grid systems · CAD plot plans · single-line electrical diagrams · equipment sizing · string impedance calculations · NEC 690 compliance and safety calculations · rack load calculations · footing calculations · licensed-engineer stamping coordination · electrical permit packages · city plan review · permit to operate · interconnection and utility coordination · custom mechanical engineered parts · contract administration · EPC documentation · AI-aided drafting · technical sales engineering

Energy & Thermal — air conditioning estimation · air flow and BTU calculations · building envelope analysis · seasonal thermal load calculations · passive solar design · solar azimuth and inclination · site shading analysis · solar thermal · phase-change materials · thermal energy storage · heat pumps · absorption and evaporative cooling · waste-to-energy · Fischer-Tropsch synthesis · syngas conditioning · hot gas flow control · anti-fouling · heat transfer · VFD and motor control · SCADA and instrumentation · commissioning

Program, Operations & Governance — program and project management · project delivery · pilot builds and deployment · procurement · vendor and contractor management · cost reduction · budget management · KPIs · cross-functional teams · operations management · continuous improvement · client relationships · regulatory compliance · inspections · policy development · internal controls · audits · risk management · litigation · open-meetings law · technical communication

EXPERIENCE

INDEPENDENT SOFTWARE ENGINEER — FREELANCE

September 2025 – Present

- Web and AI development across a portfolio of client projects, each taken from scoping through build, deployment, and support: a business system for a lapidary shop, real estate operations in Nevada County, an animal shelter project in San Quintín, and an AI literacy program. Also bid and priced solar engineering work through PhaseSave.

PHASESAVE.COM — PARTNER / DIRECTOR, RESEARCH & DEVELOPMENT

Palm Springs, CA · 2017 – 2021, Present

- Produces permit-ready solar packages in CAD: plot plans, single-line electrical diagrams, equipment sizing, string impedance calculations, and NEC 690 compliance and safety calculations. Coordinates licensed-engineer review and stamping, then runs the package through city plan review to sign-off and permit to operate. Recovered a non-permitted photovoltaic installation by working the City on a permit-to-operate backed by as-built engineering, bringing an unapproved system into compliance and back online.
- Designs and fabricates custom mechanical engineered parts where a catalog item does not fit — racking and mounting hardware and site-specific assemblies — and produces permit packages on AI-aided drafting software.
- Estimates and sizes cooling loads for air conditioning work — air flow and BTU calculations, building envelope analysis, and seasonal thermal load calculation for desert and data center applications — and develops waterless phase-change material thermal shelving for those cooling loads.
- Directs R&D and consulting delivery across sales engineering, project management, vendor sourcing, and contractor management, with depth in solar power, air conditioning, phase-change materials, cold-chain logistics, and tax credits. Brought Monobloc air-to-water heat pumps into California as sales and engineering representative.

KNOWLEDGECITY LLC — PRODUCT OWNER, INTEGRATIONS

Senior Quality Assurance Engineer → Test Automation Lead (years 2–4) → Product Owner · December 2020 – August 2025

- Owned product quality for six enterprise integrations — SAP, Oracle, Workday, Coursera, UKG, and Zoom — over SAML, OAuth, SFTP, REST, custom APIs, and webhooks, with SCIM, SCORM, and LTI extending coverage across HRIS and learning platforms.

- Built the test automation function from nothing as Test Automation Lead: recruited, hired, and trained the engineers, then built an end-to-end regression suite with health dashboards and alerting across deployment scenarios.
- Owned white-box test development against a CI/CD pipeline on Apache, REST, SQL, and AWS behind a React front end — Cypress, Selenium, Postman, JUnit, Git, Node, Elastic, Qase, Jira.
- Set the quality bar across integrations, front end, back end, mobile, security, performance, load, API, database, usability, accessibility, and localization including right-to-left Arabic. Shipped an AI support assistant with retrieval-augmented generation and guardrails under SOC 2 controls. Worked blue-team incident forensics to root cause.

EARLY RENEWABLE ENERGY COMPANIES — EWSOLAR.NET (VP), FREE ENERGY RESOURCES (CTO), GRIDNOT (VP), ONPOINT POWER (Partner), DESERT POWER, SOLATRON

Lucerne Valley, Homeland, Phelan, Norco and Palm Springs, CA · 2004 – 2016

- GRIDNOT: oversaw teams and vendors for the sales and integration of solar, wind, and geothermal air-conditioning systems, and did the passive solar design work — solar azimuth and inclination selection and site shading analysis for each installation. Sold more than \$32 million in installations.
- EWSOLAR: sold and developed large direct solar-fired VFD air-conditioning pumping projects at 30HP and above, from concept through customer approval and field deployment. Owned sales engineering and contractor management, site documentation, and commissioning.
- DESERT POWER: supported projects across PV solar, waste-to-energy, and tri-generation combined electricity, heat, and refrigeration. Researched Fischer-Tropsch syngas conversion to liquids — process stabilization, anti-fouling, hot gas flow control, and absorption refrigeration.
- ONPOINT POWER: sold and engineered residential and light-commercial photovoltaic and thermal solar. Pioneered a waste-to-energy proposal for the City of Norco in 2006, presenting at an agendized council meeting on converting horse manure at the wastewater plant into energy; Chevron subsequently built the facility. FREE ENERGY RESOURCES: built and ran a Korean solar marketing company as CTO over sales, procurement, permitting, and finance.

MARIANA RANCHOS COUNTY WATER DISTRICT — PRESIDENT, BOARD OF DIRECTORS

Apple Valley, CA · 2006 – 2010 · California special district

- Delivered a SCADA and instrumentation upgrade for tank and pump sensors and relays, adding remote telemetry and automated control. Cut operating expenditures 53% on a \$16,027,777 annual budget while explicitly protecting capital expenditure projects.
- Rebuilt district operations: replaced the General Manager, tightened purchasing and internal controls, and moved water testing and monitoring to trained staff. Governed under the Brown Act through a term including active litigation.

AMP COMPUTER AND LUCERNE VALLEY UNIFIED SCHOOL DISTRICT — FREELANCE CONSULTANT AND IT MANAGER

Apple Valley and Lucerne Valley, CA · 2002 – 2011

- Consulted and sold for eight years covering server and PC hardware, ERP, CRM, and office automation, then moved into web development with Django and WordPress, owning client relationships end to end.
- Provided district-wide IT infrastructure for four campuses across WAN and LAN, NetWare servers, and Windows, Linux, and virtualized desktops. Owned the extract, transform, and load path for state attendance reporting — data quality, database table locking, and certification testing against a fixed deadline.

REALNETWORKS — SENIOR SOFTWARE ENGINEER II, STREAMING MEDIA PLATFORM

Seattle, WA · 1999 – 2001

- Tested decode stream optimization across architectures including ARM, programming directly on Psion 9210 Communicators running EPOC and Pocket PC — flashing builds to the device, capturing crash and memory-allocation logs off-target, and reproducing playback stalls under low-bandwidth and low-memory conditions. That isolated media pipeline defects simulator-based testing had been passing, because the failure existed only on the device.
- Started on front-end SDET, then joined the advanced research team covering servers, embedded, Linux, and Unix back-end testing. Delivered presales engineering for the embedded Linux consumer electronics stack internationally with the VP of Sales.

MICROSOFT AND KEENE INC. — SOFTWARE TEST ENGINEER AND TECHNICAL SUPPORT

Redmond, WA and Seattle, WA · 1996 – 1999

- Automated nightly build test scripts and verification for Windows Media Server 4 and 5, the streaming media product line and a direct lead-in to RealNetworks. Verified OEM setup across hardware and driver combinations for Windows 98 and NT.

IBM — SENIOR TEST ENGINEER III, HIGH AVAILABILITY SERVERS

Kirkland, WA · 1998

- Drove WHQL certification testing for high-availability cluster servers by exercising RAID configurations under sustained high-load webserver traffic while injecting simulated drive failures — proving failover held under real production load, with no data or service lost across controller, array, and host failure. Designed the failure-injection and load harness it ran on.

SELECTED TECHNICAL WORK

- herdr-mcp (github.com/eugeneb50/herdr-mcp) — author. 51-tool Rust Model Context Protocol server with an Axum HTTP bridge and React playground, recipe engine, cron scheduler, and agent lifecycle observability.

EDUCATION & CREDENTIALS

- Associate of Science, **Mechanical Engineering** — Victor Valley College, 1994 – 1996
- WHQL certification testing, high-availability cluster servers — IBM, 1998
- California Notary Public — commissioned · Bilingual English and Spanish, with right-to-left Arabic localization work on production software